

TIM 172A, Lecture # 15, 11/18/25

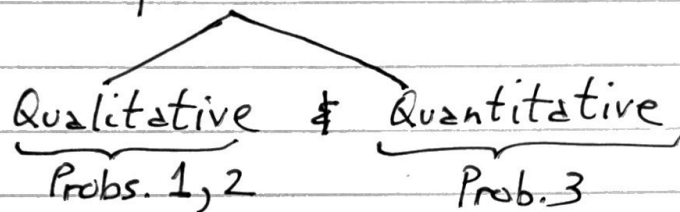
Agenda

1. HW # 7 (very short, straightforward homework)
 2. Prototyping Strategy
 3. Product Architecture
 - Product platform
 - Product Lines
- } Product family

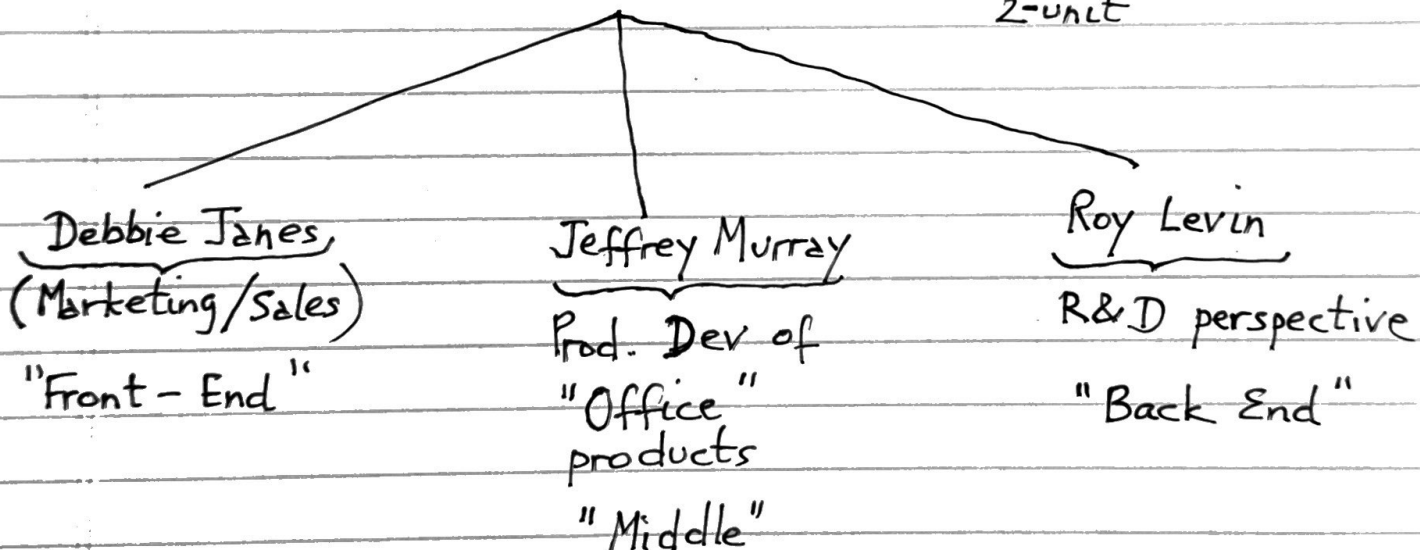
1. HW #7

- is a very short, straightforward home-work due on Thursday, 11/20/25

- 3 problems



- Prob #1 : 3 Microsoft presentations
(from previous TIM 170 presentations)
2-unit



⇒ Lessons learned from the MS presentations relative to your own work on the midterm.

Prob. # 2 : Project Management at Cisco (by Ben Rus) & Seagate (Kevin Mann)

Two methodologies

- Waterfall
- Agile



answer the straightforward questions in Prob. 2

Prob. # 3

Financial Modeling

Theory

will be covered
in L#17,18

Practice:

ability to do
complex financial
modeling in EXCEL

Baby-step : Do the
"base-case" or nominal
financial model in Excel



Prob. # 3 : Replicate the Base-case
scenario on Pages 1-7 of the Excel
Financial Modeling tutorial.

2. Prototyping (Strategy)

Use the 8-step product conceptual design process (L # 12, 13, 14) to generate 1-2 feasible concepts for further development.

The next step in the product development process is to "build & test" several prototypes for each concept $\Rightarrow$ leads to a prototyping strategy

Process for the prototyping strategy :

Objective : Build & test several types of prototypes

Step 1 : After generating a feasible concept, the product design and development team performs several "build-and-test" cycles of prototyping

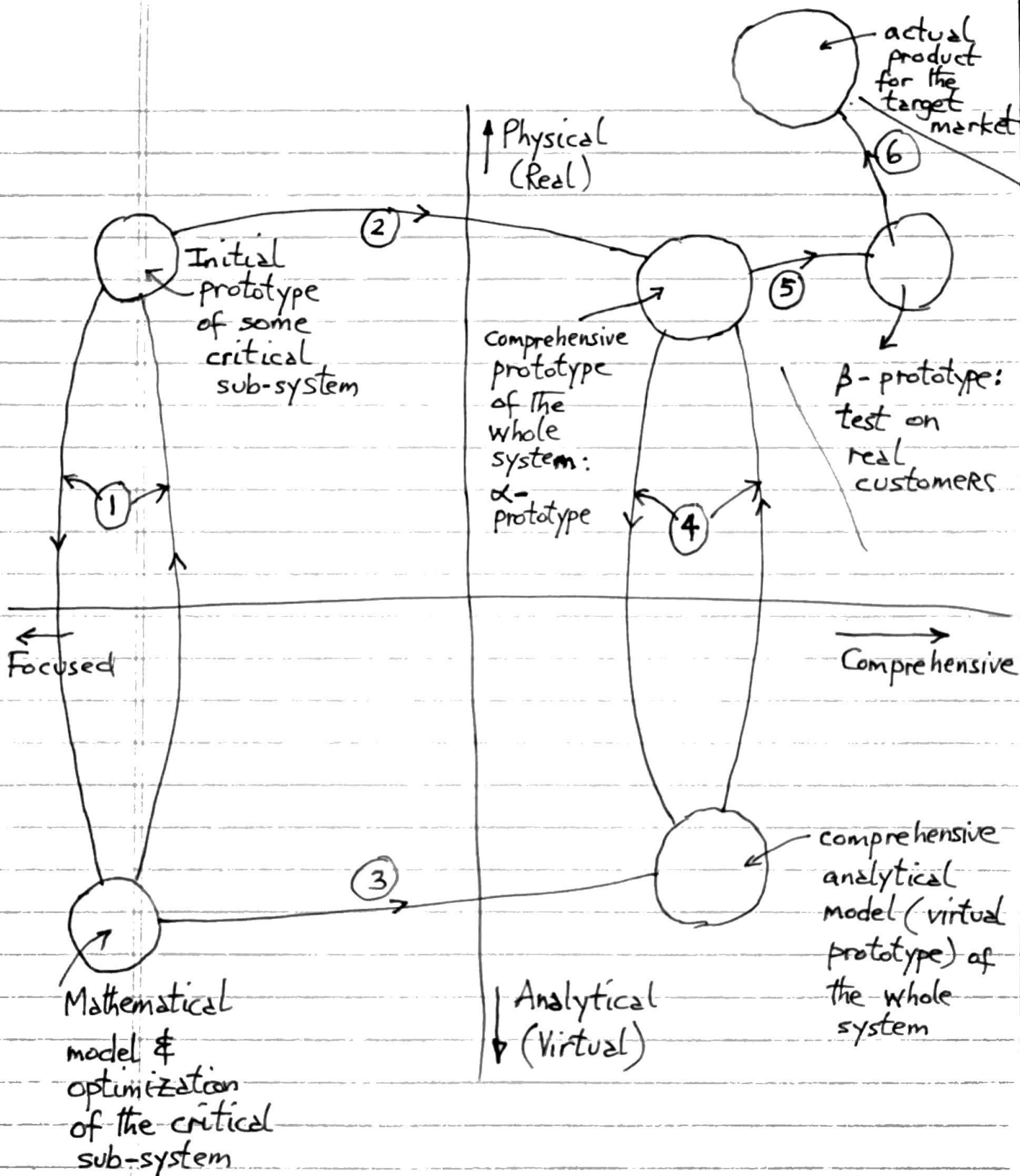
Step 2 : Define a prototyping space, & a path (or trajectory) or prototyping strategy

X-axis

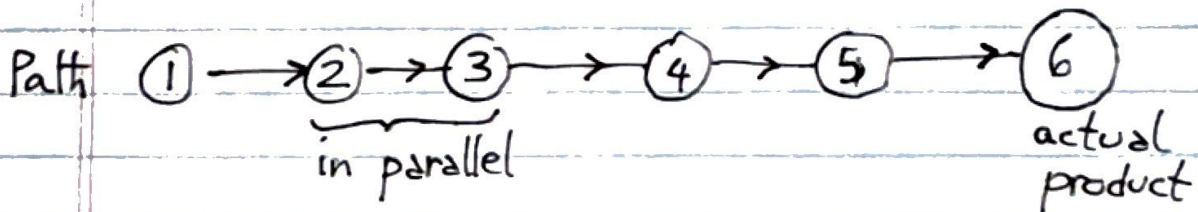
- Focused (e.g. sub-system) prototype
- Comprehensive prototype

Y-axis

- Physical (real)
- Analytical prototype
($\Rightarrow$ Virtual prototype)



Prototyping strategy in this case (fairly typical; see above)



Example :

Read the "Prototyping" chapter in the PD&D text by U&E.

product : Packbot by iRobot
product company

3. Product Architecture

So far the PD&D team has

1. completed the product conceptual design process $\Rightarrow$ 1, 2 feasible concepts.
2. built-and-tested several prototypes using an appropriate prototyping strategy

The PD&D team is now ready to build the product platform :

the infrastructure from which to develop a family of product lines to serve different market segments.

3.1 Product platform & product lines

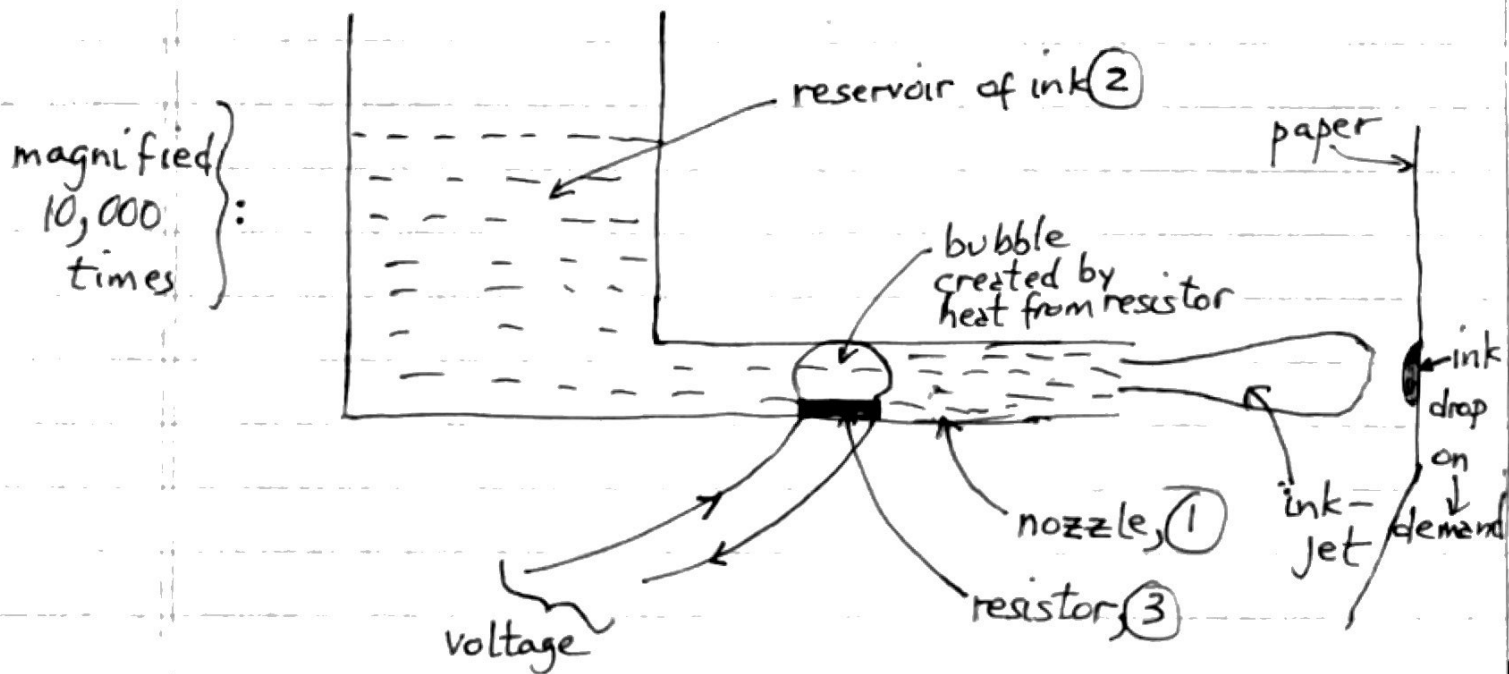
Prob. statement : How are product platforms & product lines created?

Process (or Plan or Approach) & Implementation

Step 1 : starting by creating the core or defining technology elements that are unique to the product.

Example : The Thermal Ink-Jet (Think Jet) printer

What are the core (or defining) elements of the Think-jet printer?



Core or defining technology elements:

- (1) nozzle
- (2) reservoir of ink
- (3) resistor